CHAPTER 3

MORPHOLOGY OF FLOWERING PLANTS

CHAPTER ESSENCE AT A GLANCE:
Structural diversification of vegetative organs and floral whorls driving cross-pollination, mechanical defense, and seed dispersal.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Root and Stem Adaptive Variations:** Roots adapt beyond anchorage and water uptake: tap roots of carrot store carbohydrates, prop roots of banyan provide pillar-like vertical support, stilt roots of maize arise from lower nodes, and pneumatophores of mangrove Rhizophora grow vertically upward for atmospheric respiration in saline swamps.

• **Phyllotaxy and Leaf Architecture:** Phyllotaxy represents the pattern of leaf arrangement on stems: alternate (single leaf per node in China rose), opposite (pair of leaves per node in Calotropis), and whorled (more than two leaves per node in Alstonia). Leaves modify into tendrils for climbing (pea) and protective spines (cactus).

• **Aestivation Patterns of Floral Envelopes:** Aestivation is the positional arrangement of sepals or petals in the floral bud: Valvate (margins touch without overlapping in Calotropis), Twisted (one margin overlaps adjacent petal in China rose, cotton), Imbricate (margins overlap irregularly without defined direction in Cassia, Gulmohur), and Vexillary (papilionaceous with one large standard petal, two lateral wings, and two fused keel petals in pea).

• **Placentation and Drupe Fruit Anatomy:** Placentation describes ovule arrangement within the ovary: Marginal (pea), Axile (tomato, citrus), Parietal (mustard, argemone), Free-central (Dianthus, primrose), and Basal (sunflower). The drupe is a simple fleshy fruit derived from a monocarpellary superior ovary with an outer epicarp, fleshy or fibrous mesocarp, and stony endocarp.

ESSENTIAL TAXONOMICAL FRAMEWORKS & PROCESSES

$$\text{Solanaceae Formula} = \oplus\ ¤\text{extdied}\ \text{K}_{(5)}\ \overbrace{\text{C}_{(5)}\ \text{A}_5}\ \underline{\text{G}}_{(2)}$$

Floral Formula of Family Solanaceae

$$\text{Vexillary Corolla} = 1\ \text{Standard} + 2\ \text{Wings (Alae)} + (2)\ \text{Keel (Carina)}$$

Papilionaceous Floral Architecture in Fabaceae

$$\text{Drupe Anatomy} = \text{Epicarp (Skin)} + \text{Mesocarp (Pulp or Fibre)} + \text{Endocarp (Stony)}$$

Pericarp Differentiation in Mango and Coconut

$$\text{Placentation Forms} = \text{Marginal} + \text{Axile} + \text{Parietal} + \text{Free-central} + \text{Basal}$$

Ovarian Ovule Attachment Topologies

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** In Mango, the edible portion is the sweet fleshy mesocarp; in Coconut, the mesocarp is fibrous and inedible, while the edible portion is the cellular and liquid endosperm.

• **Alert:** Do not confuse imbricate aestivation (irregular overlapping) with vexillary aestivation (distinct 1 posterior standard, 2 lateral wings, and 2 fused anterior keels).

• **Alert:** Pneumatophores are negatively geotropic respiratory roots with pneumathodes (lenticels) adapted specifically for oxygen-deficient waterlogged saline habitats.

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. Both Mango and Coconut are classified as drupe fruits. Distinguish between their mesocarps in terms of anatomical structure and economic utility. [Score: 2 Marks]

Answer & Key Valuation Steps:

• Mango drupe: Mesocarp is middle, fleshy, sweet, and edible. **[1 Mark]**

• Coconut drupe: Mesocarp is fibrous, dry, and inedible, yielding commercial coir fiber. **[1 Mark]**

2. Define phyllotaxy. Name any two distinct phyllotaxy patterns and provide one plant example for each. [Score: 2 Marks]

Answer & Key Valuation Steps:

• Definition: Phyllotaxy is the pattern of arrangement of leaves on the stem or branch. **[1 Mark]**

• Alternate phyllotaxy: Single leaf at each node (e.g., China rose, Mustard). **[0.5 Mark]**

• Opposite phyllotaxy: Pair of leaves arise at each node opposite to each other (e.g., Calotropis, Guava). **[0.5 Mark]**

3. Describe four major types of aestivation observed in the floral envelopes of angiosperms with illustrative examples. [Score: 3 Marks]

Answer & Key Valuation Steps:

• Valvate: Sepals or petals in a whorl merely touch one another at the margin without overlapping (e.g., Calotropis). **[0.75 Mark]**

• Twisted: One margin of the appendage overlaps that of the next one in a regular clockwise or anti-clockwise direction (e.g., China rose, Lady finger, Cotton). **[0.75 Mark]**

• Imbricate: Margins of sepals or petals overlap one another but not in any particular direction (e.g., Cassia, Gulmohur). **[0.75 Mark]**

• Vexillary: Largest posterior petal (standard) overlaps two lateral petals (wings), which in turn overlap two smallest anterior petals (keel) (e.g., Pea, Bean). **[0.75 Mark]**

4. Explain five distinct types of placentation observed in the ovaries of flowering plants, giving one plant example for each. [Score: 3 Marks]

Answer & Key Valuation Steps:

• Marginal: Placenta forms a ridge along the ventral suture of the ovary and ovules are borne in two rows (e.g., Pea). **[0.6 Mark]**

• Axile: Placenta is axial and ovules are attached to it in a multilocular ovary (e.g., China rose, Tomato, Lemon). **[0.6 Mark]**

• Parietal: Ovules develop on the inner wall of the ovary or on peripheral parts; often unilocular becoming two-celled by false septum (e.g., Mustard, Argemone). **[0.6 Mark]**

• Free-central: Ovules are borne on a central axis and septa are completely absent (e.g., Dianthus, Primrose). **[0.6 Mark]**

• Basal: Placenta develops at the base of the ovary and a single ovule is attached to it (e.g., Sunflower, Marigold). **[0.6 Mark]**

5. Write the complete floral formula of family Solanaceae and describe its calyx, corolla, and gynoecium characteristics. [Score: 3 Marks]

Answer & Key Valuation Steps:

• Floral formula: Actinomorphic, bisexual, Calyx 5 gamosepalous, Corolla 5 gamopetalous, Androecium 5 epipetalous, Gynoecium bicarpellary syncarpous superior ovary. **[1 Mark]**

• Calyx: Sepals 5, united (gamosepalous), persistent, valvate aestivation. **[0.75 Mark]**

• Corolla: Petals 5, united (gamopetalous), valvate aestivation; stamens 5 epipetalous. **[0.75 Mark]**

• Gynoecium: Bicarpellary, syncarpous, superior ovary, bilocular with swollen axile placenta carrying many ovules, obliquely placed carpels. **[0.5 Mark]**

6. Explain how pneumatophores assist halophytes like Rhizophora in thriving within waterlogged mangrove environments. [Score: 3 Marks]

Answer & Key Valuation Steps:

• Pneumatophores are specialized respiratory roots that grow vertically upward (negatively geotropic) out of waterlogged soil. **[1 Mark]**

• In mangrove swamps, soil is waterlogged and severely depleted of oxygen, preventing subterranean root respiration. **[1 Mark]**

• Pneumatophores possess specialized breathing pores termed pneumathodes (lenticels) that take in atmospheric oxygen directly for root metabolic survival. **[1 Mark]**

7. Contrast family Solanaceae with family Fabaceae regarding: (a) Floral symmetry, (b) Corolla aestivation, (c) Stamen arrangement and cohesion, and (d) Placentation type. [Score: 4 Marks]

Answer & Key Valuation Steps:

• (a) Floral symmetry: Solanaceae flowers are actinomorphic (radial symmetry); Fabaceae flowers are zygomorphic (bilateral symmetry). **[1 Mark]**

• (b) Corolla aestivation: Solanaceae shows valvate aestivation of united petals; Fabaceae shows vexillary (papilionaceous) aestivation of polypetalous corolla. **[1 Mark]**

• (c) Stamen arrangement: Solanaceae has 5 epipetalous stamens alternating with petals; Fabaceae has 10 stamens in diadelphous condition (9 fused plus 1 free). **[1 Mark]**

• (d) Placentation: Solanaceae exhibits axile placentation with swollen placenta in a bicarpellary ovary; Fabaceae exhibits marginal placentation in a monocarpellary unilocular ovary. **[1 Mark]**